Business Forecasting

Chapter 1:
Introduction to Forecasting

Course: BBAZ16605

Instructor: Jamal Abdul Nasir

Thanks to Prof. Yan (Jenny) Chen

A little about me

Name: Jamal Abdul Nasir

Education:

- Ph.D. City University of Hong Kong 2018
- M.S. Korea Advanced Institute of Science and Technology 2013
- B.Sc. University of the Punjab, Pakistan 2009

Professional Experience:

1. Research Assistant /Post-doctoral fellow at The University of Hong Kong 2018 –2023
2. Assistant Professor at Macau University of Science and Technology 2023 – Present

Industry Experience:

Officer Industrial Engineering at Interloop Limited, Pakistan October 2009–August 2011

Course Mechanics

- **Teaching and learning activities:**
  - Lecture + In-class debate and discussions + Class exercises+ Group project and presentations
- **Course website:**
  - https://moodle.must.edu.mo
  - Resources/Assessment materials/Course announcement
  - Lecture notes will be regularly posted before or immediately after each class
- **Textbook:**
  - Hanke, J.E., and Wichern, D.W. (Latest Edition). Business forecasting.

Office hours by appointment:

- Mon.,9:30-12:00, Tue., 9:30-12:00, Wed., 9:30 - 12:00, Thu., 9:30- 12:00

- **Communication:**
  - Email: njabdul@must.edu.mo

Evaluation Criteria

- **Attendance + Class participation** 10%
  - Individual grades
- **Assignments** 20%
  - Individual grades
  - No points will be deducted from the assignments submitted within 24h after the due date.
  - No assignment will be accepted after 24h in any case.
  - Assignments can't be submitted through emails.
- **Final exam** 50%
  - Individual grades
- **Group project** 20%
  - Presentation and report
  - Group grade

- **Total** **100%**

Group Projects Information submission

- Submit the group information (6~8 students in each group) through the class representative by 24/01/2026 using the Excel sheet and follow the format as shown below:

A	B	C	D	E	F	G
Business Forecasting_D1-2025_Groups						
Sr#	Group Name	Group Member	Student ID	Chinese Name	English name	Photo
1	**A**	1	1210			
2		2	12100			
3	**B**	1	1210			
4		2	1210			
5		3	1210			

Forecasting

- What is forecasting?
  - A common task in business to inform decision-making.
  - Often involved in predicting important variables.
    - Monthly sales of different products of Amazon.com.
    - Future interest/exchange rate fluctuation.

Forecasting

- How can an operations manager realistically set production schedules without some estimate of future sales?
- How can a company determine staffing for its call centers without some guess of the future demand for service?
- How can a bank make realistic plans without some forecast of future deposits and loan balances?

Everyone requires forecasts!

History of Forecasting

- Peter Bernstein (1996) notes that the development of business forecasting in the seventeenth century was a major innovation. He writes:

"Forecasting—long denigrated as a waste of time at best and a sin at worst—became an absolute necessity in the course of the seventeenth century for adventuresome entrepreneurs who were willing to take the risk of shaping the future according to their own design."

- Over the next 300 years, significant advances in data-based forecasting methods occurred, with much of the development coming in the twentieth century.

- Along with the development of data-based methods, the role of judgment and judgmental approaches to forecasting has grown significantly over the last 25 years.

History of Forecasting

- Without any data history, human judgment may be the only way to make predictions about the future. In cases where data are available, judgment should be used to review and perhaps modify the forecasts produced by quantitative procedures.

- Using powerful personal computers and sophisticated software packages, forecasts of future values for variables of interest are easily generated.

- The ease of computation does not replace clear thinking and experience.

Attitudes about forecasting

- Two extreme attitudes to forecasting:

- It is a waste of time because its almost impossible to get a 100% true forecast.
  - ❖ least expectation
- A forecast of very high quality can solve all of our problems.
  - ❖ Very high and unrealistic least expectation

Need to address these attitudes appropriately!

- There is always scope to improve the forecast and get more accurate results (Reliable data, forecasting method, software)
- Lack of managerial oversight and improper use of forecasting techniques can lead to costly decisions.

Forecasts in business

- Demand forecasting /Sales forecasting
  - basis for all other forecasting
- Purchasing forecast
- Distribution forecast
  - regional sales plan, shipping and arranging transportation vehicles, Warehouses, etc.
- Capacity planning
  - production planning including production machines, raw material, equipment, personnel, etc.
- Financial forecasting
  - determine expectations of future results (cost, sales, revenue)
- Product life cycle forecasting

Forecasting

- How to forecast?
  - Involve extending past experience to the future.

Categories of forecasts

- Qualitative vs. Quantitative
- Qualitative: only judgment involved, no obvious manipulation of data.
- For example,
  - Delphi method (專家調查法)
  - Panel discussion (小組討論)
  - Market Survey
  - Executive Opinion

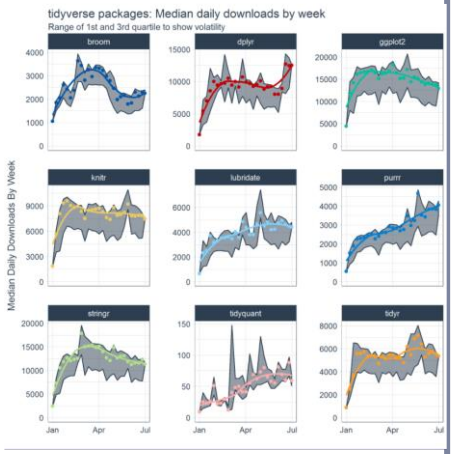
Categories of forecasts

- *Qualitative methods* are used when the situation is *vague* and *little data exists*. Typically, such situations have a long-term nature or include decisions that are to be taken about new technology or product. Forecasting for such cases involves intuition and experience.

  - The problem with the *expert methods* is that they are based on *opinions*, *experiences*, and *knowledge of human beings*, and are therefore subjective.

Categories of forecasts

- Qualitative vs. Quantitative
- Quantitative: No input of judgment, just the mechanical procedures that produce the results. For example,
  - Time series analysis
  - Regression analysis



- Must be used together in practice

❖ Judgment and common sense must be used along with mechanical and data-manipulative procedures.

Categories of forecasts

- Time horizon:

• Long-term forecasts (>5 Y)	• Strategic decisions affect and shape the direction of the whole business.
• Medium-term forecasts (1.5 – 5 Y)	• Tactical decisions help to implement the strategy.
• Short-term forecasts (<1.5 Y)	• Operational decisions relate to the day-to-day running of the business.

Forecasting Levels

Level	Horizon	Purposes
Strategic	Year/Years	• Business Planning • Capacity Planning • Investment Strategies
Tactical	Quarterly	• Brand Plans • Budgeting • Sales Planning • Manpower Planning
	Months/Weeks	• Short-term Capacity Planning • Master Planning • Inventory Planning
Operational	Days/Hours	• Transportation Planning • Production Planning

Categories of forecasts

- Micro vs. Macro
  - Small details v.s. large summary values
    - Number of lecturers of MUST v.s. Labor of Macau

- Top management would be interested in forecasting the sales of the entire company, for example, whereas individual salespersons would be much more interested in forecasting their own sales volumes.

Categories of forecasts

- Nature of the output
  - Point forecast
  - Interval forecast
  - Density forecast
- The future is never exactly like the past
- The uncertainty surrounding point forecasts suggests the usefulness of an interval forecast. However, if forecasts are solely the result of judgment, point forecasts are typically the only option.

Three axioms about forecasting

1. Forecasting errors are inevitable.

2. Aggregated forecasts are more accurate.

(a family or group of products are more accurate than demand forecasts for a single product)

3. Shorter horizon forecasts are more accurate.

Selection of a forecasting method

- The prime consideration in choosing a forecasting method is that the results must facilitate the decision-making process of the organization's managers.

- Rarely does one method work for all cases.

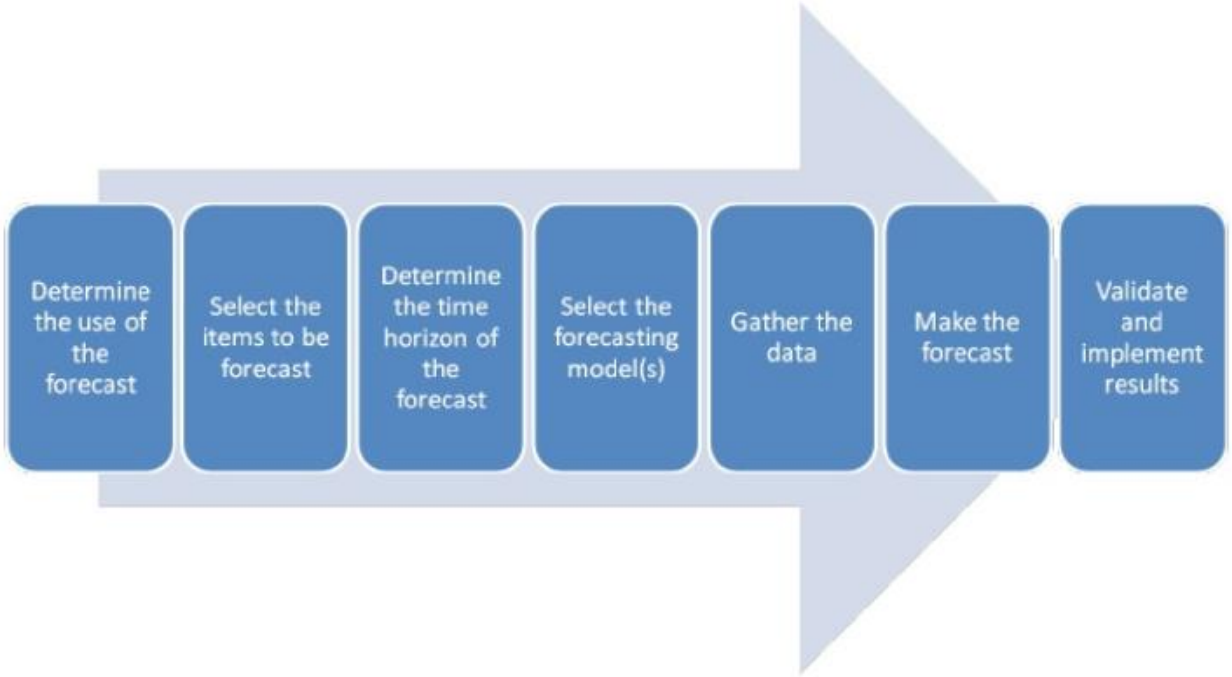
- Several factors described to categorize the forecasting need to be considered when choosing an appropriate forecasting method (point forecast or interval forecast).
  - Is the forecast needed for some point in the near future (a short-term forecast) or for a point in the distant future (a long-term forecast)?

Selection of a forecasting method

- Due to the availability of current forecasting softwares, it is best to think of forecasting methods as generic tools that can be applied simultaneously.
- Several methods can be tried in a given situation. The methodology producing the most accurate forecasts in one case may not be the best methodology in another situation.
- However, the method(s) chosen should produce a forecast that is accurate, timely, and understood by management so that the forecast can help produce better decisions.

Forecasting Steps

Seven Steps in Forecasting



Forecasting Steps

- The *forecasting process* begins with the *definition of the objective*.

- Forecasting the *number of printed materials* for a university lecture based on the prediction of *student attendance*.

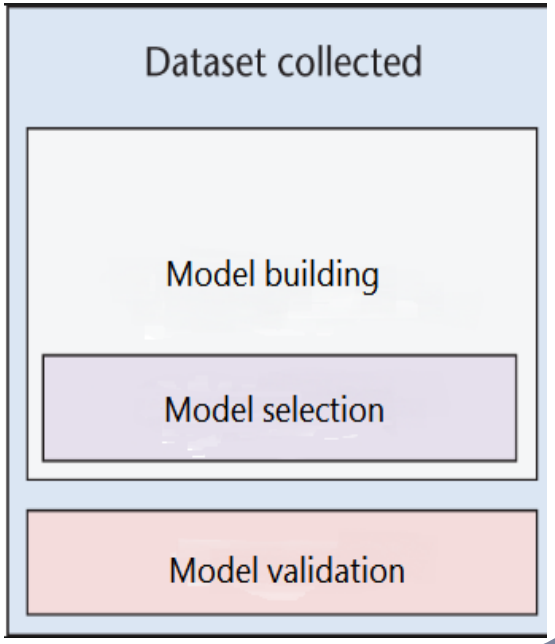
Forecasting steps

Define Objective	How many handouts should be printed fot the next lecture?
Define Horizon	Weekly
Select the Method	Time series analysis
Data Acquisition	Attendance list
Perform Forecasting	Exponential smoothing
Control	Weekly

Forecasting Steps

- The recognition that forecasting techniques operate on the data generated by historical events leads to the identification of the following five steps in the forecasting process.

- Data collection.
- Data assembling and cleaning.
- Model building and evaluation.
- Model implementation (the actual forecast)
- Forecast evaluation



Software used for forecasting

- Minitab
- SAS
- SPSS.

- There are many add-ins or supplemental programs that provide forecasting tools in a spreadsheet environment. For example, the Analysis ToolPak add-in for Microsoft Excel provides some regression analysis and smoothing capabilities.

Software used in class

- Minitab 21.1. 0
  - https://www.minitab.com/en-us/products/minitab/free-trial/

Example 1.2 Large Web-based Retailer

One of the goals of a large Internet-based retailer is to be the world's most consumer-centric company. The company recognizes that the ability to establish and maintain long-term relationships with customers and to encourage repeat visits and purchases depends, in part, on the strength of its customer service operations. For service matters that cannot be handled using website features, customer service representatives located in contact centers are available 24 hours a day to field voice calls and emails.

Because of its growing sales and its seasonality (service volume is relatively low in the summer and high near the end of the year), a challenge for the company is to appropriately staff its contact centers. The planning problem involves making decisions about hiring and training at internally managed centers and about allocating work to outsourcers based on the volume of voice calls and emails. The handling of each contact type must meet a targeted service level every week.

To make the problem even more difficult, the handling time for each voice call and email is affected by a number of contact attributes, including type of product, customer, and purchase type. These attributes are used to classify the contacts into categories: in this case, one "primary" category and seven "specialty" categories. Specific skill sets are needed to resolve the different kinds of issues that arise in the various categories. Since hiring and training require a 6-week lead time, forecasts of service contacts are necessary in order to have the required number of service representatives available 24 hours a day, 7 days a week throughout the year.

Pat Niebuhr and his team are responsible for developing a global staffing plan for the contact centers. His initial challenge is to forecast contacts for the primary and specialty categories. Pat must work with monthly forecasts of total orders (which, in turn, are derived from monthly revenue forecasts) and contacts per order (CPO) numbers supplied by the finance department. Pat recognizes that contacts are given by

$$\text{Contacts} = \text{Orders} \times \text{CPO}$$

For staff planning purposes, Pat must have forecasts of contacts on a weekly basis. Fortunately, there is a history of actual orders, actual contacts, actual contacts per order, and other relevant information, in some cases, recorded by day of the week. This history is organized in a spreadsheet. Pat is considering using this historical information to develop the forecasts he needs.

END OF CHAPTER 1